

Sebaceous Excess and Scalp Expression in the Sialic-Glycan Inflammatory Cascade

Dandruff as a Hypothesized Regional-Expansion Signal

Hamer Glycan Systems

hamerglycans.com

SGIC Patient Outcomes Registry — registry.hamerglycans.com

STATUS OF THIS PAPER

This is a hypothesis-generating paper. It does not claim to establish that dietary Neu5Gc or wheat-derived ATIs/gliadins cause scalp flaking. It proposes a mechanistic pathway that is biologically plausible, consistent with a long-term self-tracking participant's observation, and testable — and it identifies exactly what data would be needed to move the hypothesis from plausible to supported. Readers, clinicians, and researchers are invited to challenge, replicate, or falsify what follows.

1. Introduction

The SGIC framework, developed over more than two decades of self-directed observation and cross-referenced against the sialic acid biology literature (notably the work of the Varki laboratory at UC San Diego on Neu5Gc), proposes that dietary Neu5Gc and wheat-derived amylase-trypsin inhibitors (ATIs) and gliadins drive a chronic, low-grade inflammatory cascade. Prior papers in this series have argued this cascade manifests visibly through sebaceous gland hyperactivity, presenting clinically as acne and excess facial oiliness.

This paper extends that argument to a tissue not previously covered in the series: the scalp. It asks whether dandruff — specifically, the pattern of excess oiliness and visible flaking — is a downstream expression of the same sebaceous mechanism already proposed for acne, appearing at a different anatomical site with its own local variables (yeast ecology, hair as a collection medium) layered on top.

It also raises a second, more speculative question, prompted by the participant's own recollection: whether early scalp involvement functions as a regional-expansion signal — an early indicator that the same underlying process might later spread to, or intensify at, other sebaceous-rich sites (neck, upper back, shoulders) as the condition progresses with age. This is presented explicitly as a hypothesis about disease trajectory, not an established pattern.

2. Background: Normal Epidermal Turnover

All human skin, scalp included, undergoes continuous renewal. Keratinocytes are produced in the basal layer, differentiate as they migrate upward, and are shed from the stratum corneum in a cycle lasting approximately 28 days in healthy adult skin. This shedding is normally invisible — cells are lost individually or in microscopic clusters, unnoticed in ordinary hygiene and grooming.

This baseline process is universal and is not itself pathological. Any discussion of “excess” flaking must be understood against this backdrop: some sloughing is simply normal physiology occurring everywhere on the body, all the time, in every person. The question this paper addresses is not why skin sheds, but why the scalp, in a subset of individuals, sheds visibly and in a manner associated with oiliness — and whether that “why” connects to the same mechanism proposed elsewhere in the SGIC series.

3. Why the Scalp Is a High-Sensitivity Site

Three independent factors converge to make the scalp a site where an underlying sebaceous excess would be expected to show up more visibly than elsewhere on the body, even if the excess itself is systemic rather than scalp-specific.

3.1 Sebaceous Gland Density

The scalp carries one of the highest concentrations of sebaceous glands on the body, comparable to the face and upper trunk — the same regions implicated in acne within the existing SGIC model. If a systemic driver increases sebum output generally, the scalp is anatomically positioned to show that excess prominently, for the same structural reason the face does.

3.2 Malassezia Ecology

Dermatology research converges on a three-factor model of dandruff and seborrheic dermatitis: sebum availability, *Malassezia* metabolic activity, and individual susceptibility to the irritant byproducts produced. Increased sebum production, inherited individual susceptibility, and irritation from microorganisms such as *Malassezia* are the recognized triggering factors, with dandruff affecting more than half the world's population. Two species, *M. globosa* and *M. restricta*, are consistently identified as the predominant organisms on dandruff-affected scalp, with *M. globosa* considered the most likely initiating organism due to its high lipase activity.

The mechanistic link to sebum is direct rather than incidental: whole-genome sequencing of *M. globosa* revealed the genome lacks a fatty acid synthase gene, a deficiency not seen in other free-living fungi, which likely explains why *Malassezia* species depend on external fatty acids for growth. Unable to synthesize its own lipids, the organism is obligately dependent on sebum as an external fatty-acid source — meaning its metabolic activity should scale directly with how much sebum substrate is available to it. Its secreted lipases hydrolyze sebum triglycerides, and experimental work has shown *Malassezia* activity increases free fatty acid levels while lowering triglyceride levels on the scalp, with excess fatty acids penetrating the stratum corneum and disrupting the skin barrier.

Oleic acid, a specific free fatty acid byproduct of this process, has been directly tested as a causal agent: researchers used oleic acid as a representative *Malassezia*-produced metabolite to characterize scalp response, finding it induced flaking in susceptible individuals but not in non-susceptible individuals. This susceptibility-dependent response is consistent with the “individual susceptibility” arm of the three-factor model — the same fatty acid exposure does not produce dandruff in everyone, which may itself be relevant to why SGIC-protocol status could plausibly modulate an individual's threshold. Independent support for the sebum-*Malassezia*-flaking chain comes from treatment data: antifungal agents including zinc pyrithione, ketoconazole, selenium sulfide, and ciclopirox olamine reduce scalp *Malassezia* populations, and this reduction correlates closely with reduction in dandruff symptoms.

Taken together, this literature establishes a well-supported chain: sebum availability → *Malassezia* lipase activity → free fatty acid release → stratum corneum irritation and accelerated desquamation, gated by individual susceptibility. This is a sebum-dependent process — more available lipid substrate supports greater yeast metabolic activity, and by extension, greater irritant byproduct production. Nothing in this established chain requires any scalp-specific novel mechanism; it requires only that something upstream

increases sebum availability, which is precisely the claim already made elsewhere in the SGIC series regarding facial sebaceous glands.

3.3 Hair as a Collection and Display Medium

Flaking elsewhere on the body sheds away unnoticed. On the scalp, flakes are caught in hair, accumulate, and are dislodged visibly during grooming or overnight, against pillows and bedding. This does not change the underlying biology, but it changes what becomes noticeable — a moderate increase in flaking at the scalp is far more likely to register as a “condition” than the same relative increase elsewhere on the body.

Together, these three factors argue that the scalp is not a uniquely vulnerable tissue in some novel sense — it is an already sebum-dense, yeast-colonized, hair-covered site where an existing systemic driver would be expected to produce a disproportionately visible signal.

4. The Proposed SGIC Extension

The core SGIC claim, established elsewhere in this series, is that dietary Neu5Gc and wheat ATIs/gliadins drive sebaceous gland hyperactivity through chronic low-grade inflammatory signaling. This paper proposes extending that claim, with appropriate caution, in two tiers.

4.1 Primary Hypothesis — Sebum-Substrate Pathway (Moderate Confidence)

If SGIC-driven sebaceous hyperactivity is systemic rather than confined to facial skin, the scalp should show a parallel increase in sebum output. Increased sebum availability would be expected to support increased *Malassezia* metabolic activity and irritant byproduct production, plausibly manifesting as increased flaking severity — without requiring any scalp-specific mechanism beyond what is already proposed for acne. This is the more defensible of the two hypotheses in this paper, because it requires no new biological claim, only an extension of an existing one to a second sebaceous-rich tissue.

4.2 Secondary Hypothesis — Siglec-Mediated Immune Tolerance (Lower Confidence)

Prior work in this series (the parasite/drug-resistance whitepaper) proposed that dietary Neu5Gc and wheat ATIs may drive Siglec-mediated immune tolerance, reducing immune surveillance of certain pathogens. It is possible, though considerably more speculative, that reduced local immune surveillance could permit greater *Malassezia* overgrowth independent of sebum substrate alone. This hypothesis is flagged explicitly as weaker and less supported than 4.1, and is included for completeness rather than as a claim the participant currently stands behind with confidence.

4.3 What Is Not Being Claimed

This paper does not claim Neu5Gc or ATIs act directly on scalp tissue, nor that dandruff is evidence of a novel disease mechanism distinct from the sebaceous pathway already proposed for acne. The claim is

narrower: the same known driver, expressed at a second sebum-rich site, with known local variables (yeast, hair) layered on top.

5. Observational Case: A Long-Term Self-Tracking Participant

The hypotheses in this paper originate from the multi-decade self-observation of a long-term self-tracking participant within the SGIC framework, reported here as an illustrative single case (n=1), not as evidence of causation.

During a period of severe, scarring facial and truncal acne accompanied by excessive facial and scalp oiliness, this participant experienced substantial dandruff — heavy flaking associated with greasy hair, requiring frequent washing. Following dietary elimination of Neu5Gc- and ATI/gliadin-rich foods (the same dietary change associated with resolution of the participant's acne), the dandruff resolved in parallel with the acne and the facial oiliness, without any separate antifungal or scalp-targeted intervention.

Two observations from this case are relevant to the hypotheses above:

- The resolution occurred as a byproduct of dietary change directed at acne, not as a result of any scalp-specific treatment. This is more consistent with a shared upstream cause (sebum excess) than with an independent scalp condition that happened to co-occur.
- On reflection regarding timing, scalp symptoms did not precede acne in a strict chronological sense — they appeared at the same time as, or shortly after, acne onset in adolescence. However, the participant's recollection is that the appearance of scalp involvement carried a subjective quality of an early signal — a sense, at the time and in retrospect, that scalp involvement might represent an early marker of a process that could later expand to, or intensify at, other sebaceous-rich regions (neck, upper back, shoulders) as the underlying condition progressed with age. This is reported as a subjective impression, not a documented chronological sequence, and is treated in this paper as a hypothesis about disease trajectory rather than an established finding.

This case does not establish causation. It is the observation that generated the hypotheses in Section 4, and it is presented here in that spirit — as the origin of a question, not the answer to it.

6. The Regional-Expansion Hypothesis

A further question, raised directly by the participant's recollection above, is whether early scalp involvement might function as an early signal of a broader process — one that later manifests, or intensifies, across other sebaceous-rich regions of the body (neck, upper back, shoulders) as an individual ages.

Three non-exclusive explanations are worth naming for why this might occur, if it occurs at all:

Progressive sebaceous change with age. Sebaceous gland activity itself changes over the lifespan (typically increasing through puberty and young adulthood, and gradually declining later in life), and if SGIC-driven hyperactivity is superimposed on this natural curve, the pattern of visible expression could plausibly shift across body regions over time as the two curves interact.

Changes in local antifungal/immune defense. If the Siglec-tolerance mechanism in Section 4.2 is real, its effects might not be uniform across the body or across the lifespan, and a gradual decline in local immune surveillance at a given site could allow that site to “activate” as a visible expression point later than others.

Cumulative or threshold effects. It is possible that sebaceous excess must cross a site-specific threshold before producing visible symptoms, and different regions may cross that threshold at different ages depending on local gland density, hair coverage, and yeast colonization history.

None of these explanations is currently supported by data beyond the single case in Section 5. They are offered as candidate mechanisms for a pattern the participant has observed subjectively, worth testing rather than asserting.

7. What Would Be Needed to Test This

Moving these hypotheses from plausible to supported requires data beyond a single case. The following steps are proposed, several of which can be implemented directly through the existing SGIC Patient Outcomes Registry infrastructure.

Registry field addition. Add scalp oiliness and flaking severity as tracked fields, using a validated or adapted scale (e.g., an adherent-scalp-flaking severity score), alongside the existing acne severity tracking already collected.

Onset-order data. Systematically capture, across participants, whether scalp symptoms appeared before, concurrent with, or after facial acne onset. This is the direct test of the regional-expansion hypothesis in Section 6 — if scalp involvement reliably precedes or coincides with later expansion to other regions across a meaningful number of participants, that is a materially stronger claim than the participant's single subjective recollection.

Provocation-remission pairs. Among participants who cycle on and off the SGIC dietary protocol (voluntarily or inadvertently), track whether scalp oiliness and flaking rise and fall in parallel with acne severity. This is the strongest evidence tier available from self-report data, since it tests within-person co-variation rather than relying on cross-sectional comparison.

Age-stratified expansion tracking. For participants with longer histories, capture whether involvement of additional regions (neck, upper back, shoulders) correlates with age, disease duration, or specific life periods of dietary non-adherence.

Explicit acknowledgment of what is not measured. No *Malassezia* quantification (fungal load, species-level colonization) is proposed in this data collection plan. This paper is about sebum substrate availability, not fungal density, and readers should understand that as a real limitation of the evidence base rather than an oversight.

8. Limitations

No fungal quantification. *Malassezia* density and species composition are not measured in the proposed dataset or in the participant's case. The mechanism in Section 4.1 is inferred from established dermatological literature on sebum-*Malassezia* interaction, not from direct measurement in SGIC participants.

Single-case origin. The hypotheses in this paper originate from one person's multi-decade self-observation. Self-report is subject to recall bias and motivated reasoning, particularly from a participant already invested in the broader SGIC framework.

Indirect mechanistic distance. The proposed pathway from dietary Neu5Gc/ATI exposure to visible scalp flaking requires two inferential steps (systemic sebaceous hyperactivity, then site-specific *Malassezia* interaction) rather than a single direct mechanism. No scalp-tissue biomarker data currently exists to shorten that inferential distance.

The regional-expansion hypothesis is the weakest claim in this paper. It rests on a subjective recollection of timing and significance from a single case, not on documented chronological data. It is included because it is a genuine and interesting question, not because current evidence supports it.

Alternative explanations must be considered and not dismissed, including independent seborrheic dermatitis unrelated to diet, non-dietary *Malassezia* sensitivity, climate and humidity effects, and changes in hair or scalp product use over time — any of which could account for the participant's personal resolution timeline independent of, or alongside, dietary change.

9. Conclusion

This paper proposes that dandruff — specifically excess oiliness and visible flaking of the scalp — may be a downstream expression of the same sebaceous hyperactivity mechanism already proposed elsewhere in the SGIC series for acne, rather than an independent condition. It further proposes, with considerably more caution, that early scalp involvement may serve as a signal of a broader process that later expands to other sebaceous-rich body regions with age.

Both hypotheses are testable using the existing SGIC Patient Outcomes Registry with modest additions to current data collection. Until that data exists, this paper should be read as an invitation to test a plausible idea — not as a claim that the idea is established. Other long-term followers of the SGIC dietary protocol

are encouraged to report their own scalp symptom history, onset timing relative to acne, and any regional expansion experienced with age, so that a genuine dataset — rather than a single case — can inform whether this hypothesis holds up.

Hamer Glycan Systems | hamerglycans.com | SGIC Patient Outcomes Registry: registry.hamerglycans.com